Raio **Chea**

Software Developer ·

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Education

University of Washington *Seattle, Washington*

Pursuing B.S. in Computer Science: GPA 3.71 *Sept. 2023 - Jun. 2027 (Anticipated)*

- Direct admit to Computer Science major
- Relevant Coursework: Distributed Systems, Database Management System Internals, Systems Programming, Algorithms, Data Structures, Machine Learning, Quantum Computation, Software & Design Implementation, Linear Algebra, Multi-variable Calculus

Skills

Languages	Typescript, Java, Javascript, HTML, CSS, Python, C++, C, SQL, Dart
Frameworks	React, Flutter
Cloud & Tools	AWS (Lambda, S3, EventBridge, CloudFront), Git, Unity, OpenAI API, Photon Fusion

Experience and Leadership

Software Design & Implementation Teaching Assistant *Typescript, React, HTML, CSS, Java*

June. 2025 - present *University of Washington, WA*

- Mentored over 1,000 students in software engineering best practices, helping them design, debug, and test large TypeScript and Java applications.
- Guided students in building full-stack web applications using TypeScript, React, and Java, covering client-server communication, state management, and software design best practices.
- Implemented a Java autograder that automated grading and feedback generation, improving consistency and reducing manual grading effort.

Personal Projects

contrAI *AWS Lambda, React, S3, EventBridge, CloudFront*

- Architected a fully serverless, event-driven data pipeline on AWS using EventBridge, Lambda, S3, and CloudFront to automatically process NOAA weather forecasts every six hours.
- Implemented automated ingestion and prediction workflows that transformed atmospheric forecast data into GeoJSON overlays for an interactive web mapping application.
- Developed a React interface that visualizes contrail forecasts between airports and communicates estimated climate impact in an intuitive, consumer-friendly format.

Sharded Fault-Tolerant Key/Value Service *Java*

- Developed a fault-tolerant distributed key/value store with dynamic shard rebalancing using Paxos replication, guaranteeing linearizable operations during live cluster reconfiguration.
- Engineered a distributed storage layer that automatically migrates shards between replica groups while preserving consistency through Paxos replicated reconfiguration logs and AMO request deduplication.
- Added multi-key transaction support using two-phase commit and locking to provide serializable, deadlock-free execution across shards.

SimpleDB *Java*

- Produced a relational database engine from scratch, implementing SQL parsing, logical-to-physical query planning, and an iterator-based execution engine supporting scans, joins, filtering, aggregation, and projection.
- Integrated a concurrent storage engine featuring page-level locking, LRU buffer management, transaction logging, deadlock detection, and crash recovery using redo/undo logging for quick data retrieval and multithreading.
- Implemented a cost-based query optimizer using table statistics and histograms to improve join ordering and execution plan selection.

Multithreaded Socket Web Server *C++*

- Built a multithreaded HTTP server in C++ using POSIX sockets to concurrently process client requests and serve file search operations.
- Optimized concurrent request handling using low-level networking and multithreading techniques to improve throughput under multiple simultaneous connections.

Networked Immersive VR Barista Simulation *Unity, Meta Interaction SDK, Photon Fusion, C#*

- Developed a multiplayer VR training simulation in Unity that enables users to collaboratively operate a virtual café in real time.
- Built the multiplayer networking layer using Photon Fusion, synchronizing object ownership and player interactions through state authority and RPCs in a shared virtual environment.
- Integrated OpenAI and Meta Voice SDKs to create AI-driven virtual customers capable of natural language conversations with distinct personalities.
- Optimized networking and real-time physics systems to maintain smooth multiplayer gameplay in an immersive VR environment.